



VINUNIVERSITY

Term Project Final Report

College of Engineering and Computer Science,
COMP1020 Object-Oriented Programming & Data Structures Spring 2026

Objectives

The final report is the concluding component of the term project and aims to evaluate the completeness, technical quality, and overall outcome of the developed system. The report must demonstrate the effective application of Object-Oriented Programming (OOP) principles, data structure concepts and algorithms in Java. Students are required to showcase the final implementation, system architecture, technical decisions, evaluation results, and reflections on the development process.

Guidelines

Format: The final report is limited to a maximum of **10 pages** (inclusive of all figures, diagrams, and tables) and must include the following sections:

1. ***Introduction and Project Overview.*** Provide an overview of the project, including:
 - Problem statement and motivation.
 - Project objectives and scope.
 - Key features and functionalities.
 - Any modifications made since the proposal or interim stage.
2. ***System Requirements and Specifications.*** Describe the functional and non-functional requirements of the system:
 - Core functionalities provided by the application.
 - User requirements and expected behaviors.
 - Performance, usability, or scalability considerations.
3. ***System Design and Architecture.*** Explain the overall system design and software architecture:
 - High-level architecture and module organization.
 - Diagrams (class diagrams, sequence diagrams, etc.).
 - OOP concepts, main classes, their responsibilities, and interactions.
4. ***Data Structures and Algorithms.*** Discuss the data structures and algorithms used in the project:
 - Selected data structures and reasons for choosing them.

- Time and space complexity considerations.
- Algorithms implemented and their roles in the system.
- Trade-offs and optimization decisions.

5. **Implementation Details.** Describe the implementation process and technologies used:

- Programming language, libraries, and frameworks.
- Important implementation details.
- File structure and package organization.
- External tools or APIs used (if applicable).

6. **Testing and Evaluation.** Present evidence that the system functions correctly:

- Test cases and testing methodology.
- Screenshots or sample outputs.
- Performance evaluation or benchmarking (if applicable).
- Discussion of correctness, robustness, and usability.

7. **Challenges and Solutions.** Discuss significant challenges encountered during development:

- Technical difficulties.
- Design or architectural issues.
- Team collaboration and task coordination.
- Solutions implemented and lessons learned.

8. **Conclusion and Limitations.** Summarize the final outcome of the project:

- Overall achievements and completed objectives.
- Strengths of the developed system.
- Current limitations or unresolved issues.
- Possible future improvements or extensions.

9. **Team Contributions.** Clearly describe the responsibilities and contributions of each team member throughout the project development process.

10. **References.** Include all references used in the project, such as documentation, research papers, online resources, libraries, and APIs. Follow a consistent citation format. *Note: The References section does not count toward the page limit.*

11. **Appendix.** Include supporting materials if necessary:

- Additional screenshots.
- Extended test cases.

Note: The Appendix section does not count toward the page limit.

Note: The final report must accurately reflect the completed state of the project. Teams are expected to provide clear technical evidence of implementation, demonstrate the application of OOP and data structure concepts, and present a coherent evaluation of the developed system. Reports should emphasize both technical depth and software engineering quality.

Important Dates

- **Final Report:** Deadline: **11:59 PM, June 2, 2026.**